

Lecture#08

Joint Probability Distribution

If two or more random variables are given that are defined on the same probability space, the joint probability distribution is the corresponding probability distribution on all possible combinations of outputs.

Or simply, Joint probability distribution shows probability distribution for two or more random variables.

Joint probability distribution for discrete random variables:

Suppose that a given experiment involves two discrete variables X and Y . Then the joint probability distribution of X and Y can be expressed as

$$f(x, y) = P(X = x, Y = y),$$

Which has the following properties:

1. $f(x, y) \geq 0$ for all (x, y)
2. $\sum x \sum y f(x, y) = 1$
3. $P[(X, Y) \in A] = \sum \sum_{(x, y) \in A} f(x, y)$ for any region A in the XY plain.

Example: A coin is tossed three times. If X denotes the number of heads and Y denotes the number of tails in the last two tosses, then find the joint probability distribution of X and Y .

Solution: The outcomes of the experiment and the associated probabilities are shown below:

Outcome	X	Y	$P(X, Y)$
HHH	3	0	1/8
HHT	2	1	1/8
HTH	2	1	1/8
HTT	1	2	1/8
THH	2	0	1/8
THT	1	1	1/8
TTH	1	1	1/8
TTT	0	2	1/8

It is easy to see that X assumes values 0, 1, 2, and 3, while Y assumes values 0, 1, and 2. The joint probability distribution can be written as:

X					
values					
Y values	0	1	2	3	Row sum
0	0	0	1/8	1/8	2/8
1	0	2/8	2/8	0	4/8
2	1/8	1/8	0	0	2/8
Column sum	1/8	3/8	3/8	1/8	1

Joint distribution for continuous variables:

Let X and Y be two continuous random variables. Then the function $f(x, y)$ is called the joint probability density function of X and Y if

1. $f(x, y) \geq 0$, for all (x, y)
2. $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$
3. $P[(X, Y) \in A] = \int_A f(x, y) dx dy$ for any region A in the XY plane.

The Cumulative distribution function (*cdf*) for $f(x, y)$ is defined by

$$F(x, y) = P[X \leq x, Y \leq y] = \int_{-\infty}^x \int_{-\infty}^y f(x, y) dx dy$$

The *cdf* has the following properties:

1. $0 \leq F(x, y) \leq 1$
2. $\frac{\partial^2 F(x, y)}{\partial x \partial y} = f(x, y)$, whenever F is differentiable
3. $F(x, -\infty) = F(-\infty, y) = 0$
4. $F(\infty, \infty) = 1$.

Example: Let X and Y have the following distribution

$$f(x, y) = x^2 + \frac{xy}{3}, \quad 0 \leq x \leq 1, \quad 0 \leq y \leq 2$$

$$= 0, \quad \text{elsewhere}$$

Check that $f(x, y)$ is a density function.

Solution: The function has the joint density function if

$$f(x, y) \geq 0, \quad \text{for all } (x, y)$$

And

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$$

Clearly, $f(x, y) \geq 0$, for all values of x and y in the given range. And

$$\begin{aligned} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy &= \int_0^1 \int_0^2 \left(x^2 + \frac{xy}{3} \right) dx dy \\ &= \int_0^2 \left(\frac{1}{3} + \frac{y}{6} \right) dy = \left[\frac{y}{3} + \frac{y^2}{12} \right]_0^2 = 1 \end{aligned}$$

Hence the proof.

Example: Consider the following density function:

$$\begin{aligned} f(x, y) &= \frac{x(1 + 3y^2)}{4}, \quad 0 \leq x \leq 2, \quad 0 \leq y \leq 1 \\ &= 0, \quad \text{elsewhere} \end{aligned}$$

Find $P \left[0 < X < 1, \frac{1}{4} < Y < \frac{1}{2} \right]$.

Solution: The joint probability is obtained as follows:

$$\begin{aligned} P \left[0 < X < 1, \frac{1}{4} < Y < \frac{1}{2} \right] &= \int_{1/4}^{1/2} \int_0^1 \frac{x(1 + 3y^2)}{4} dx dy = \int_{1/2}^{1/4} \left(\frac{x^2}{8} + \frac{3x^2 y^2}{8} \right)_{x=0}^{x=1} dy \\ &= \int_{1/2}^{1/4} \left(\frac{1}{8} + \frac{3y^2}{8} \right) dy = \left[\frac{y}{8} + \frac{y^3}{8} \right]_{1/2}^{1/4} = \frac{23}{512} \end{aligned}$$

Marginal Distribution:

When the distribution of the random variable (Say, X or Y) is derived from a joint probability distribution (say, $f(x, y)$), then the resulting distribution is known as a marginal distribution (of X or Y).

When the random variable X and Y are discrete, the marginal distribution of X is

$$g(x) = \sum_y f(x, y).$$

And marginal distribution of Y is

$$h(y) = \sum_x f(x, y).$$

When X and Y are random continuous variables,

$$g(x) = \int_{-\infty}^{\infty} f(x, y) dy, \quad \text{for } -\infty < x < \infty$$

$$h(y) = \int_{-\infty}^{\infty} f(x, y) dx, \quad \text{for } -\infty < y < \infty.$$

$$* \int_{-\infty}^{\infty} g(x) dx = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$$

$$\begin{aligned} P[a < X < b] &= P[a < X < b, -\infty < Y < \infty] \\ &= \int_a^b \int_{-\infty}^{\infty} f(x, y) dx dy = \int_a^b g(x) dx \end{aligned}$$

Question:

*Suppose that X and Y have the following joint probability distribution

Values of X					
Values of Y	0	1	2	3	Row sum
0	1	1/8	2/8	1/8	4/8
1	1/8	2/8	1/8	0	4/8
Column sum	1/8	3/8	3/8	1/8	1

Find the marginal distribution of X and Y .

Solution:

For random variables of X

$$g(0) = P(X = 0) = \sum_{y=0}^1 f(0, y) = f(0, 0) + f(0, 1) = 0 + \frac{1}{8} = \frac{1}{8}$$

$$g(1) = P(X = 1) = \sum_{y=0}^1 f(1, y) = f(1, 0) + f(1, 1) = \frac{1}{8} + \frac{2}{8} = \frac{3}{8}$$

$$g(2) = P(X = 2) = \sum_{y=0}^1 f(2, y) = f(2, 0) + f(2, 1) = \frac{2}{8} + \frac{1}{8} = \frac{3}{8}$$

$$g(3) = P(X = 3) = \sum_{y=0}^1 f(3, y) = f(3, 0) + f(3, 1) = \frac{1}{8} + 0 = \frac{1}{8}$$

Similarly, for Y

$$\begin{aligned} h(0) &= P(Y = 0) \\ &= \sum_{x=0}^3 f(x, 0) = f(0, 0) + f(1, 0) + f(2, 0) + f(3, 0) \\ &= 0 + \frac{1}{8} + \frac{2}{8} + \frac{1}{8} = \frac{4}{8} = \frac{1}{2} \end{aligned}$$

$$\begin{aligned} h(1) &= P(Y = 1) \\ &= \sum_{x=0}^3 f(x, 1) = f(0, 1) + f(1, 1) + f(2, 1) + f(3, 1) \\ &= \frac{1}{8} + \frac{2}{8} + \frac{1}{8} + 0 = \frac{4}{8} = \frac{1}{2} \end{aligned}$$

Marginal distribution of X :

x	0	1	2	3	Sum
$f(x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$	1

Marginal distribution of Y :

y	0	1	Sum
$f(y)$	4/8	4/8	1

This shows that the marginal distributions are probability distribution.

*Find the marginal densities of X and Y from the following density function and verify that marginal distributions are also probability distributions.

$$f(x, y) = \left\{ \frac{1}{8}(6 - x - y) \right\}, \text{ for } 0 < x < 2, 2 < y < 4.$$

Also compute $P[X + Y < 3]$ and $P[X < 1.5, Y < 2.5]$.

Solution: The marginal density of X is

$$g(x) = \frac{1}{8} \int_2^4 (6 - x - y) dy = \frac{1}{8} \left[6y - xy - \frac{y^2}{2} \right]_2^4 = \frac{1}{4}(3 - x), 0 < x <$$

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The marginal density of Y is

$$h(y) = \frac{1}{8} \int_0^2 (6 - x - y) dx = \frac{1}{8} \left[6x - \frac{x^2}{2} - xy \right]_0^2 = \frac{1}{4}(5 - y), 2 < y <$$

4.

Now, we verify that $g(x)$ and $h(y)$ are probability distributions.

It is clear that in the given range of X and Y ,

$$g(x) \geq 0 \text{ and } h(y) \geq 0$$

Also

$$\int_0^2 g(x) dx = \frac{1}{4} \int_0^2 (3 - x) dx = \frac{1}{4} \left[3x - \frac{x^2}{2} \right]_0^2 = 1$$

And

$$\int_2^4 h(y) dy = \frac{1}{4} \int_2^4 (5 - y) dy = \frac{1}{4} \left[5y - \frac{y^2}{2} \right]_2^4 = 1$$

Here $g(x)$ and $h(y)$ satisfy all the condition of a density function.

Now

$$\begin{aligned} P(X + Y < 3) &= \frac{1}{8} \int_0^2 \int_2^{3-x} (6 - x - y) dy dx = \frac{1}{8} \int_0^2 \left[6y - xy - \frac{y^2}{2} \right]_2^{3-x} dx \\ &= \frac{1}{8} \int_0^2 \left(\frac{x^2}{2} - 4x + \frac{7}{2} \right) dx \\ &= \frac{1}{8} \left[\frac{x^3}{6} - 2x^2 + \frac{7x^2}{2} \right]_0^2 = \frac{1}{24} \end{aligned}$$

$$P\left(X < \frac{3}{2}, Y < \frac{5}{2}\right) = \frac{1}{8} \int_0^{\frac{3}{2}} \int_2^{\frac{5}{2}} (6 - x - y) dy dx = \frac{9}{32}.$$

Conditional Distribution:

The conditional distributions are exactly analogous to the conditional probabilities of the type $P(A|B)$ or $P(B|A)$, where A and B are two events in a sample space. Using the definition of conditional probability

$$P(B|A) = \frac{P(A \cap B)}{P(A)}, \quad P(A) > 0$$

Replacing the events A and B by the random variables X and Y respectively, we define the conditional probability of Y for given X as follows:

$$P(Y = y|X = x) = \frac{P(X = x, Y = y)}{P(X = x)} = \frac{f(x, y)}{g(x)}$$

When X and Y are discrete random variables,

$$f(y|x) = \frac{f(x, y)}{\sum_y f(x, y)} = \frac{f(x, y)}{g(x)}, \text{ for } g(x) > 0$$

$$f(x|y) = \frac{f(x, y)}{\sum_x f(x, y)} = \frac{f(x, y)}{h(y)}, \text{ for } h(y) > 0$$

When X and Y are the continuous random variables,

$$f(y|x) = \frac{f(x, y)}{\int_{-\infty}^{\infty} f(x, y) dy} = \frac{f(x, y)}{g(x)}, \quad g(x) > 0$$

$$f(x|y) = \frac{f(x, y)}{\int_{-\infty}^{\infty} f(x, y) dx} = \frac{f(x, y)}{h(y)}, \quad h(y) > 0.$$

If one wished to find the probability that the random variable X falls between a and b , when it is known that variable $Y = y$, we evaluate

$$P(a < X < b|Y = y) = \begin{cases} \sum_x f(x|y), & \text{if } X \text{ is discrete} \\ \int_a^b f(x|y), & \text{if } X \text{ is continuous} \end{cases}$$

Example:

Values of X

Values of Y	0	1	2
0	3/28	3/28	3/28
1	6/28	6/28	0
2	1/28	0	0

Find $f(x|1)$, $f(y|1)$ and $P(X = 0|Y = 1)$.

Solution: $f(x|1) = \frac{f(x,1)}{h(1)}$ and $f(y|1) = \frac{f(1,y)}{g(1)}$.

Now, $h(1) = \sum_{x=0}^2 f(x,1) = f(0,1) + f(1,1) + f(2,1) = \frac{6}{28} + \frac{6}{28} + 0 = \frac{3}{7}$.

Hence, the conditional distribution of X given Y = 1 is

$$f(x|1) = \frac{f(x,1)}{h(1)} = \frac{7}{3} f(x,1), \text{ for } x = 0, 1, 2.$$

Therefore,

$$f(0|1) = \frac{7}{3} f(0,1) = \frac{7}{3} \times \frac{6}{28} = \frac{1}{2}$$

$$f(1|1) = \frac{7}{3} f(1,1) = \frac{7}{3} \times \frac{6}{28} = \frac{1}{2}$$

$$f(2|1) = \frac{7}{3} f(2,1) = \frac{7}{3} \times 0 = 0$$

Hence, the conditional distribution of X given Y = 1 is,

X	0	1	2
$f(x 1)$	1/2	1/2	0

Example: Given the following joint density function of X and Y.

$$f(x,y) = \frac{6}{5}(x+y^2), \quad 0 < x < 1, \quad 0 < y < 1$$

$$= 0, \quad \text{elsewhere}$$

Find the following:

- (i) $f(x|y)$
- (ii) $f(x|0.5)$
- (iii) $P(X < 0.5|Y = 0.5)$

Solution:

$$h(y) = \frac{6}{5} \int_0^1 (x + y^2) dx = \frac{6}{5} \left(\frac{x^2}{2} + xy^2 \right) \Big|_0^1$$

$$= \frac{6}{5} \left(\frac{1}{2} + y^2 \right), \quad 0 < y < 1$$

(i)

$$f(x|y) = \frac{f(x,y)}{h(y)} = \frac{\frac{6}{5}(x + y^2)}{\frac{6}{5}\left(\frac{1}{2} + y^2\right)} = \frac{2(x + y^2)}{1 + 2y^2}, \quad 0 < x < 1, \quad 0 < y < 1$$

(ii)

From $f(x|y)$ above,

$$f(x|0.5) = \frac{2(x + 0.5^2)}{1 + 2(0.5)^2} = \frac{1}{3}(4x + 1), \quad 0 < x < 1$$

(iii)

$$P(X < 0.5 | Y = 0.5) = \frac{1}{3} \int_0^{0.5} (4x + 1) dx$$

$$= \frac{1}{3} (2x^2 + x) \Big|_0^{0.5} = \frac{1}{3}.$$

Independence of Random Variables:

Two random variables X and Y with marginal densities $g(x)$ and $h(y)$, respectively, are said to be independent if and only if

$$f(x|y) = g(x) \text{ or } f(y|x) = h(y)$$

Where $f(x|y)$ is the conditional density of X for given Y .

If X and Y are independent, then

$$f(x, y) = g(x) \cdot h(y)$$

For all values of x and y .

Values of X

Values of Y	x_1	x_2	x_3	Row Sum
y_1	$f(x_1, y_1)$	$f(x_2, y_1)$	$f(x_3, y_1)$	$h(y_1)$
y_2	$f(x_1, y_2)$	$f(x_2, y_2)$	$f(x_3, y_2)$	$h(y_2)$
Column Sum	$g(x_1)$	$g(x_2)$	$g(x_3)$	1

For independence,

$$f(x_1, y_1) = g(x_1) \cdot h(y_1)$$

$$f(x_2, y_1) = g(x_2).h(y_1)$$

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$$f(x_3, y_2) = g(x_3).h(y_2)$$

Example: Suppose X and Y have the following joint probability function:

Values of X

Values of Y	2	4	Row Sum
1	0.10	0.15	0.25
3	0.20	0.30	0.50
5	0.10	0.15	0.25
Column Sum	0.40	0.60	1

Check if X and Y are independent.

Solution:

i) $f(2,1) = 0.10$ and $g(2) = 0.40, h(1) = 0.25$, hence $g(2).h(1) = 0.10 = f(2,1)$.

ii) $f(4,1) = 0.15$ and $g(4) = 0.60, h(1) = 0.25$, hence $g(4).h(1) = 0.15 = f(4,1)$.

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vi) $f(4,5) = 0.15$ and $g(4) = 0.60, h(5) = 0.25$, hence $g(4).h(5) = 0.15 = f(4,5)$.

For all points of the values (x, y) of the random variables X and Y , $f(x, y) = g(x)h(y)$

Hence, the variables are independent.

Alternatively, we can use the concept of conditional probability in this case.

$$p(Y = 1|X = 2) = \frac{f(2,1)}{g(2)} = \frac{0.10}{0.40} = 0.25 = h(1)$$

$$p(Y = 3|X = 2) = \frac{f(2,3)}{g(2)} = \frac{0.20}{0.40} = 0.50 = h(3)$$

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$$p(Y = 5|X = 4) = f(5|4) = \frac{f(4,5)}{g(4)} = \frac{0.15}{0.60} = 0.25 = h(5)$$

Here, the conditional distributions of Y for all X's are equal to the marginal distribution of Y. So, X and Y are independent.

Example:

$$f(x, y) = \frac{x + y}{8}, 0 < x < 2, 0 < y < 2$$

Show that X and Y are not independent.

Solution:

$$g(x) = \frac{1}{8} \int_0^2 (x + y) dy = \frac{x + y}{4}$$

$$h(y) = \frac{1}{8} \int_0^2 (x + y) dx = \frac{y + 1}{4}$$

Thus, $g(x)h(y) = \frac{(x+1)(y+1)}{4} \neq f(x, y)$.

Hence, X and Y are not independent.

Alternatively, using the concept of conditional distribution,

$$f(y|x) = \frac{f(x, y)}{g(x)} = \frac{\frac{x + y}{8}}{\frac{x + 1}{4}} = \frac{x + y}{2(x + 1)} \neq h(y)$$

$$f(x|y) = \frac{f(x, y)}{h(y)} = \frac{\frac{x + y}{8}}{\frac{y + 1}{4}} = \frac{x + y}{2(y + 1)} \neq g(x)$$

Since the conditional distributions are not equal to the marginal distributions, the variables are not independent.